

COS80022 – Software Quality and Testing

Report for Lab 1

Student Name: AJULA PEEDIKAPARAMBIL RAJU

Student ID:105286562

Student email: 105286562@student.swin.edu.au

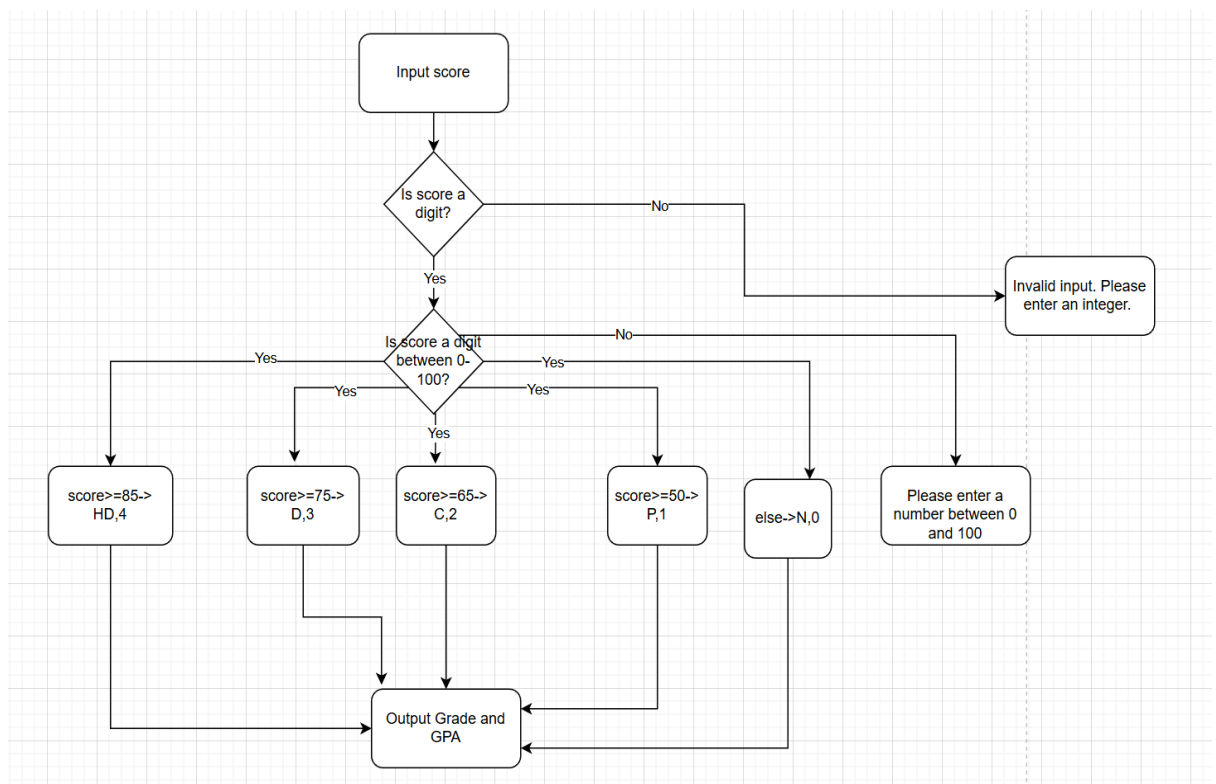
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1. User Requirements

This program accepts a positive integer between 0 and 100, representing the final mark a student gets for an academic unit, and outputs the corresponding grading code (HD, D, C, P, N) and GPA value (4, 3, 2, 1, 0).

2. Analysis and Design

If the input is an integer between 0 and 100, the program assigns a grade and GPA according to University's grading scheme. The program should follow the workflow below:



3. Coding

A python code is created and the source code is as follows:

```
1
2 # Python Program to findout the grade and GPA from mark
3
4 score = input("Enter your final mark (0-100): ")
5
6 if score.isdigit():
7     score = int(score)
8     if 0 <= score <= 100:
9         if score >= 85:
10             grade, gpa = "HD", 4
11         elif score >= 75:
12             grade, gpa = "D", 3
13         elif score >= 65:
14             grade, gpa = "C", 2
15         elif score >= 50:
16             grade, gpa = "P", 1
17         else:
18             grade, gpa = "N", 0
19         print(f"Grade: {grade}, GPA: {gpa}")
20     else:
21         print("Please enter a number between 0 and 100.")
22 else:
23     print("Invalid input. Please enter an integer.")
24
```

4. Testing

The following test cases were executed:

#	Input Data	Expected Result
TC1	85	HD, GPA: 4
TC2	75	D, GPA: 3
TC3	65	C, GPA: 2
TC4	50	P, GPA: 1
TC5	45	N, GPA: 0
TC6	101	"Please enter a number between 0 and 100."
TC7	A	"Invalid input. Please enter an integer."

The execution results of the test cases are shown as follows:

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): 85
Grade: HD, GPA: 4
```

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): 75
Grade: D, GPA: 3
```

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): 65
Grade: C, GPA: 2
```

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): 50
Grade: P, GPA: 1
```

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): 45
Grade: N, GPA: 0
```

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): 101
Please enter a number between 0 and 100.
```

```
C:\SoftwareT\venv\Scripts\python.exe C:\SoftwareT\integer.py
Enter your final mark (0-100): A
Invalid input. Please enter an integer.
```

The actual execution results are as expected. The program has passed all the test cases